Argentina Oil & Gas: Vaca Muerta

Vaca Muerta has doubled Argentina's oil output since 2017 on rock that rivals the Permian. New pipelines and LNG exports arrive in 2027, and Milei's reforms let companies take new profits home. The question is whether the reforms outlast the 2027 election.

Research note · Draft as of November 13, 2026

Introduction

Argentina is the only country outside North America where shale oil works at scale. Vaca Muerta, a formation in northern Patagonia about the size of Belgium, now produces seven of every ten Argentine barrels and has pushed national output to a record of nearly one million barrels a day, double the level of 2017. The rock is not the constraint: its best wells out-produce good Permian wells foot for foot, though they cost about 30% more to drill. What held the play back sat above ground, in full pipelines, capital controls that trapped profits and governments that set prices. All three are changing at once. A new oil export pipeline to the Atlantic starts up in early 2027, the main gas pipeline grows by two-thirds a few weeks later and large-scale LNG exports begin by year-end. The partners in the far larger Argentina LNG project have set November 26 for their final go-ahead. President Milei's investment regime, RIGI, offers large projects thirty years of tax and currency stability, and foreign companies can again send new profits home, though older earnings stay trapped. YPF, the state-controlled champion, operates half the basin's oil, and most global majors have sold to local operators while Chevron expands. The weak point is the macro. The economy is contracting, country risk is elevated and Milei is tied in the polls with Axel Kicillof, the Peronist who designed the 2012 seizure of YPF.

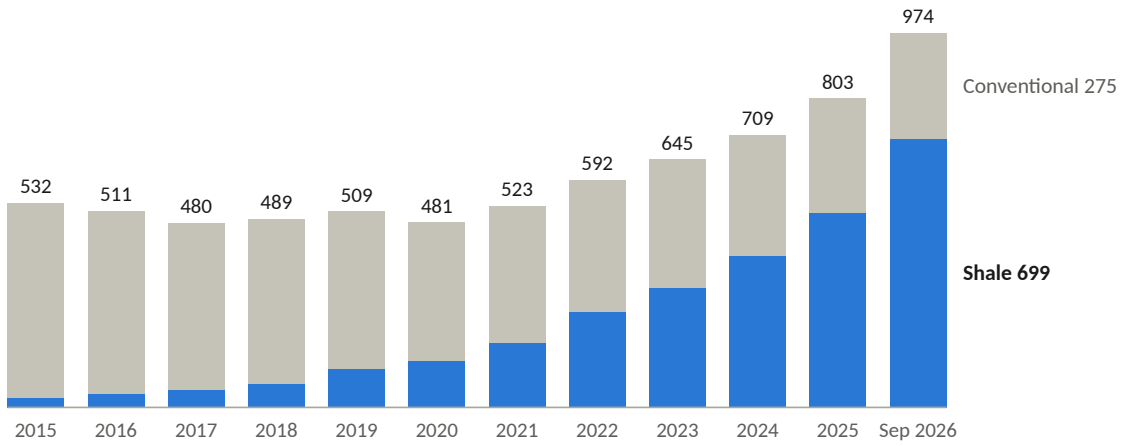
Key Points

- Vaca Muerta supplies about 70% of Argentina's oil. National output hit a record 974,000 barrels a day in September, double the 2017 level.
- The bottleneck is pipelines, not geology. The main oil export line has been full since July 2026.
- Relief arrives in 2027: a new oil pipeline (first cargo in March or April), a larger gas pipeline (by May) and the first of two floating LNG vessels (year-end).
- The next catalyst is November 26, when YPF, Eni and Abu Dhabi's XRG are due to give the final go-ahead to Argentina LNG.
- RIGI gives large projects a 25% tax rate, relief from export duties after three years and thirty years of stability. About US$76bn of projects are approved, nearly all in oil, gas and mining.
- Foreign companies can take cash out again, but only profits earned from 2025 onward.
- Shale crude still pays an 8% export duty whenever Brent is above US$60, the main fiscal irritant left for producers.
- The best wells beat good Permian wells foot for foot but cost about 30% more to drill.

- YPF operates about half the basin's oil. ExxonMobil, Petronas and Equinor have sold to Pluspetrol and Vista, while Chevron has filed a US$13.8bn plan to grow.
- The main risk is political. Milei is tied for 2027 with Axel Kicillof, who designed the 2012 expropriation of YPF.

Shale has doubled Argentine oil output since 2017

Crude oil production, thousand barrels a day. Annual averages; the last bar is the month of September 2026.



Source: Energy Secretariat data as reported by El Cronista (totals 2021-2025, shale 2025), elDiarioAR (2019 total, shale 2019-2020), Yahoo Noticias (shale 2021-2024), and Ámbito and Bloomberg Línea (September 2026). Totals for 2015-2018 are from a third-party data series and 2020 from GlobalData. Shale before 2019 is estimated from Los Andes, El Economista and iProfesional; 2017 is interpolated. Conventional is total minus shale.

Conventional output has fallen every year since 2015. All of the growth is shale.

Fiscal Regime

RIGI is the centerpiece, and it is working mostly for oil and gas. The Incentive Regime for Large Investments, known by its Spanish acronym RIGI, is Milei's flagship investment law. A qualifying project pays corporate tax at 25% instead of 35%, imports equipment free of duty, stops paying export duties after three years and can keep a rising share of its export dollars offshore. Those terms are frozen for thirty years and backed by international arbitration. About US$76bn of projects have been approved, almost all in oil, gas and mining, and the energy list reads like a map of Vaca Muerta: the VMOS oil pipeline, the Southern Energy LNG project, TGS's gas pipeline expansion and upstream developments by Pampa, Pluspetrol and Tecpetrol. The biggest applications are still pending, led by Argentina LNG, a US$25bn YPF oil development and Chevron's El Trapial plan. The window to apply closes in July 2027.

Outside RIGI, Milei freed prices but kept the tax that hurts most. His 2024 reform law ended the government's power to set domestic crude prices, so producers now sell at international prices and export freely. The exception is the export duty. Shale crude pays the full 8% whenever Brent is above US$60, while older conventional fields were given relief this year. The Plan Gas fixed-price supply contracts end in 2028 and will not be renewed, so gas producers need export markets to replace them. The provinces own the resource and collect royalties, which in Neuquén are 12% on older licenses and 18% on new ones.

Foreign companies can get cash out, with one important limit. Capital controls, known as the cepo, blocked dividends for about ten of the past fifteen years. Since April 2025 the central bank has let companies remit profits earned from fiscal 2025 onward, and they have used the channel heavily: about US$4bn left in the first nine months of 2026, the most in a decade, with energy companies leading. Profits earned before 2025 are still stuck, and holders were offered a dollar bond from the central bank instead. This freedom rests on a central bank rule that a new government could reverse overnight. RIGI projects are better protected, because their currency rights are written into the law itself.

The macro is the weak leg, and the 2027 election decides it. Inflation has fallen from triple digits to the low 30s and the IMF program remains in place. But the economy is contracting, poverty has risen again to 32% and country risk spiked above 900 basis points in early November. Milei's party won the October 2025 midterms yet lacks a majority in either chamber. Polls put him level for 2027 with Axel Kicillof, the Peronist governor of Buenos Aires province, who as deputy economy minister designed the 2012 expropriation of YPF. RIGI is the industry's hedge against that outcome, not a guarantee: Argentina has broken long-term promises to investors before.

Midstream, Pipelines and LNG

Today there is one main way out for oil, and it is full. About three-quarters of the basin's crude travels east on the Oldelval pipeline to the Atlantic port of Puerto Rosales, near Bahía Blanca. Oldelval is owned by the producers themselves, led by YPF and Pluspetrol. An expansion completed in April 2025 more than doubled it to about 540,000 barrels a day, and it was full again by July 2026. A smaller line, OTASA, carries about 110,000 barrels a day over the Andes to a refinery in Chile, and the overflow moves by truck at much higher cost. Argentina's refineries are running near capacity, so almost every additional barrel has to be exported and pipeline space sets the pace of growth.

VMOS adds three-quarters more capacity, with the first cargo due in early 2027. Vaca Muerta Oil Sur is a new 437 km pipeline from the Allen hub to a deep-water terminal at Punta Colorada on the Río Negro coast, the first in Argentina able to load the largest crude tankers. It costs about US$3bn, 70% of it funded by an international bank loan, and is owned by the companies that will ship on it, led by YPF with roughly 30%. Construction is nearly finished. Oil starts filling the line in January 2027, the first cargo is expected in March or April, and capacity builds to 550,000 barrels a day by the end of 2027. A second project, Oldelval's Duplicar Norte, connects the newer northern blocks to the system.

The projects under way buy the basin room to about 2030. Oldelval's chief executive says the works in progress lift the basin's capacity from about 720,000 barrels a day today to more than 1.3 million. The industry's goal is 1.5 million barrels a day by 2030. Getting there needs a further round of projects: more pumping on VMOS, which could take it past 700,000 barrels a day, and another line from the northern blocks. BBVA Research puts the infrastructure still needed across oil and gas at US$15-20bn by 2030.

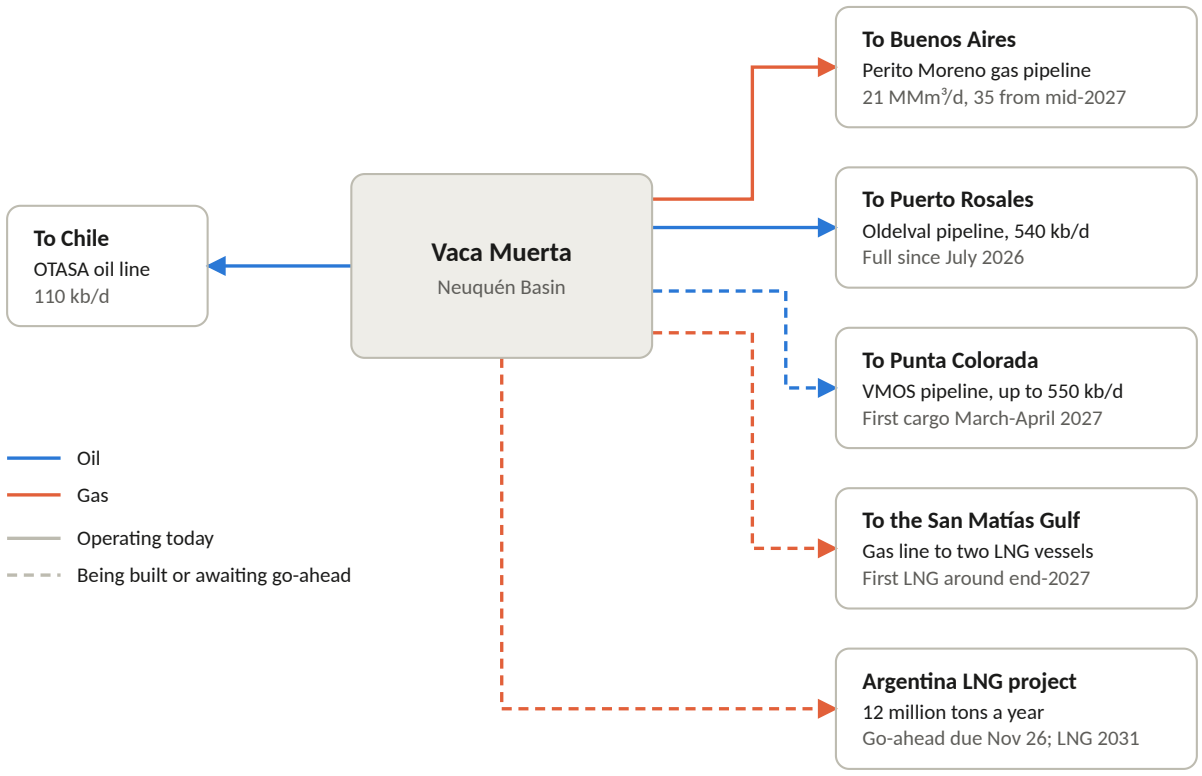
Gas has the opposite problem: plenty of supply, not enough ways to sell it. Vaca Muerta can produce far more gas than Argentina uses in summer, yet the country still imports LNG every winter because heating demand spikes and there is no storage. The main artery is the Perito Moreno pipeline, built by the state in 2023

and operated by TGS. TGS is spending its own money to expand it by two-thirds, from 21 to 35 million cubic meters a day (0.7 to 1.2 bcf/d), in time for winter 2027, and the new space was heavily oversubscribed. Gas also flows to Chile, and in small volumes to Brazil through Bolivia.

LNG is how Argentina plans to turn stranded gas into exports. Two projects matter. Southern Energy, led by Pan American Energy with YPF, Pampa, Harbour and Golar LNG, has already committed to two floating liquefaction vessels off the Río Negro coast. The first starts up around the end of 2027 and the second in 2028. Together they are modest, about 6 million tons a year, but they are fully sanctioned. The larger prize is Argentina LNG, a YPF-led venture with Italy's Eni and XRG, the international arm of Abu Dhabi's ADNOC. It is twice the size, needs about US$29bn before first LNG in 2031 and lost Shell as a partner in late 2025. YPF says the final go-ahead will be signed on November 26. Financing and gas sales contracts are still open, so the signature starts the test rather than ending it.

One oil route to the coast today; a second, and LNG, from 2027

How Vaca Muerta oil and gas leave the basin. Schematic, not to scale.



Oil has one main exit to the coast today. VMOS adds a second in 2027, and the two LNG projects give gas its first large export outlet.

Who owns what

Asset	Capacity	Main owners	Status
Oldelval oil pipeline to the Atlantic (Puerto Rosales)	540 kb/d	YPF 37%, Pluspetrol 33%, Chevron 14%, Pan American 12%, Pampa 2%, Tecpetrol 2%	Operating, full since July 2026
OTASA oil pipeline to Chile	110 kb/d	ENAP, Chile's state oil company, 36%; YPF 36%; Chevron 28%	Operating
VMOS oil pipeline and tanker terminal (Punta Colorada)	180 kb/d at start-up, 550 kb/d by end-2027	YPF 30%, Pluspetrol 15%, Chevron 12%, Pan American 10%, Pampa 9%, Vista 9%, Shell 7%, Tecpetrol 7%, Neuquén's GyP 1%	First cargo March or April 2027
Perito Moreno gas pipeline	21 MMm³/d today, 35 after expansion (0.7 to 1.2 bcf/d)	Owned by Enarsa, the state energy company; operated and expanded by TGS	Expansion due by May 2027
Southern Energy LNG	About 6 million tons a year, with a dedicated gas pipeline	Pan American 30%, YPF 25%, Pampa 20%, Harbour 15%, Golar 10%	First vessel around end-2027, second in 2028
Argentina LNG	12 million tons a year	YPF 36%, Eni 32%, XRG 32% (upstream venture, pending closing)	Final go-ahead due November 26; first LNG 2031

Source: Oldelval: Argenports, Global Energy Monitor, La Nación. OTASA: LM Neuquén, Global Energy Monitor. VMOS: ADNSUR, VMOS credit rating report, Shale24, S&P Global. Perito Moreno: TGS, EconoJournal. Southern Energy: Offshore Technology, Oil & Gas Journal, Shale24. Argentina LNG: Eni, YPF 6-K, Shale24, Ámbito.

Vaca Muerta: Geology and Operations

Vaca Muerta sits in the Neuquén Basin of northern Patagonia, about 1,000 km southwest of Buenos Aires. The formation covers roughly 30,000 square kilometers, about the size of Belgium. Nearly all of today's production comes from Neuquén province, though the formation extends into Mendoza, Río Negro and La Pampa. Activity is concentrated around the town of Añelo, about 100 km northwest of Neuquén city.

It is one of the largest shale resources on earth, and barely touched. The US Energy Information Administration put it at 308 trillion cubic feet of recoverable gas and 16 billion barrels of oil, enough to rank Argentina second in the world for shale gas and fourth for shale oil. An Argentine study this year nearly doubled the oil figure, to 30 billion barrels. Only about 5% has been developed. The limit on growth is capital and infrastructure, not drilling inventory.

The play is divided into windows by what the rock produces. The shale gets deeper and hotter moving west, so the east produces oil, the west dry gas and the middle a mix rich in condensate. Nearly all the money today goes into the oil window around Añelo. Four blocks there, Loma Campana, La Amarga Chica, Bandurria Sur and Bajada del Palo Oeste, produce just under half the basin's oil. The gas window to the west, anchored by Tecpetrol's Fortín de Piedra, is being drilled less while it waits for pipelines and LNG. A newer "northern hub" of oil blocks, including Chevron's El Trapial, has good rock but higher costs because of distance.

Vaca Muerta is one formation among several, and it is the one that matters. It is a marine shale laid down around the end of the Jurassic, the source rock that fed the basin's older conventional fields. It lies about 2,500 to 3,500 meters down (8,000 to 11,500 feet), above the Tordillo sandstone and below the Quintuco

limestone. What sets it apart is thickness. The productive section can be several hundred meters thick, far more than most US shales, which lets operators land wells at several levels in the same spot. It is also naturally overpressured, which helps wells flow. Other formations in the basin, notably the deeper Los Molles shale and the Lajas tight gas sands, are known targets but see far less activity.

Drilling started in 2010 and became an industrial operation over the past decade. YPF, then controlled by Repsol, drilled the discovery well in 2010 and brought in Chevron in 2013 to develop Loma Campana, still narrowly the largest field. More than 3,300 wells have been drilled since, almost all in Neuquén and almost all horizontal since 2016. Activity is at a record, with 43 rigs running in September.

A typical well runs two miles sideways and costs US$12-14m. Operators drill down about 3,000 meters, then sideways for about 3,300 meters (10,800 feet), and fracture the rock in 50 to 60 stages. Wells are drilled in groups from a single pad and stacked at two to five depths, and drilling takes about two weeks. The longest laterals now exceed 5,000 meters. The cost is about 30% above a comparable Permian well, by Chevron's estimate, because rigs, sand and services are scarcer and travel further, and in the northern hub a well can cost US$20m. The payoff is productivity. Rystad puts first-year output at about 30 barrels per lateral foot, against 15 to 23 in the Permian, and Vista says its drilling inventory breaks even below US$45 a barrel.

The Players

YPF is the anchor of the basin. The state-controlled company operates about half of Vaca Muerta's oil, owns the largest stakes in the pipelines and leads Argentina LNG. It is selling its old conventional fields to concentrate on shale. For any other operator, YPF is likely to be a partner, a shipper on the same pipe, or both.

Behind it sits a group of fast-growing local companies. Vista Energy is the pure-play shale oil producer and the most active acquirer. Pan American Energy, Pluspetrol, Pampa Energía and Tecpetrol, part of the Techint group that also owns Tenaris, round out the domestic field. These companies have been the buyers as international majors leave.

The majors have mostly been sellers, with Chevron the clear exception. ExxonMobil sold to Pluspetrol, and Petronas and Equinor sold to Vista. Shell was reported in January to be sounding out buyers, which its chief executive denied. Chevron is going the other way, with a US$13.8bn, thirty-year plan for El Trapial. TotalEnergies has sold oil acreage to YPF but says it has close to US$10bn still to invest in Argentina, mostly in gas. New US money is appearing too: Continental Resources has a preliminary deal to take half of Phoenix Global Resources, a mid-sized shale oil producer.

The companies at a glance

Company	What it does in Argentina	Argentine production	Spending
YPF	The national oil company, 51% state-owned. Operates about half the basin's oil, leads the pipelines and Argentina LNG, and is shedding conventional fields to become a shale company.	532 kboe/d, including 224 kb/d of shale oil (Q3 2026)	US$5.8-6.2bn in 2026
Vista Energy	A pure Vaca Muerta oil company founded by former YPF chief Miguel Galuccio. Grew by buying out Petronas and Equinor, and exports about three-quarters of its oil.	169 kboe/d (Q3 2026)	US$460m in Q3
Chevron	YPF's original shale partner at Loma Campana and sole owner of El Trapial. Wants to triple Argentine output to 195 kboe/d by 2035. Shareholder in VMOS, Oldelval and the Chile pipeline.	65 kboe/d net (2025)	US$13.8bn over 30 years, pending RIGI approval
Harbour Energy	A non-operating gas partner, through assets bought from Wintershall Dea. Most of its production is offshore Tierra del Fuego, not Vaca Muerta. Its shale exposure is minority stakes in two TotalEnergies-operated blocks plus 15% of Southern Energy LNG.	74 kboe/d (Jan-Sep 2026)	Not disclosed for Argentina
Pampa Energía	A power and gas company building a shale oil business at its wholly owned Rincón de Aranda block, which it aims to double to 45 kb/d by 2028. Holds 20% of Southern Energy LNG and 9% of VMOS, and co-controls TGS.	21 kb/d of oil at Rincón de Aranda (Q3 2026)	US$585m in Q3 company-wide, US$158m of it at Rincón de Aranda
TGS	Argentina's largest gas pipeline company. Operates the Perito Moreno line, is expanding it at its own cost and is building a US$3bn gas liquids project for 2030. Co-controlled by Pampa.		More than US$500m in 2026
Tenaris	Makes the steel pipe that lines wells, at its Campana mill, and has built the basin's third-largest fracking business. Part of the Techint group.		US$110m on its newest frac fleet
Halliburton	The basin's largest fracking contractor, with about 60% of stages in October, and YPF's main completions provider under a multi-year contract signed in April. Bringing the first electric frac fleet to Argentina.		
Helmerich & Payne	US drilling contractor expanding fast. Runs about a quarter of the basin's rigs today and expects 15 working by August 2027, about a third.		

Source: Company third-quarter 2026 results and releases, and press reports. YPF: 6-K and call. Vista: call and Equinor. Chevron: Reuters, Oil & Gas Journal and, for 2025 production, The Rio Times. Harbour: trading update. Pampa: call and slides. TGS: call, release and MDZ. Tenaris: release. Halliburton: release and Petroleumag. Helmerich & Payne: results and call. YPF's share of basin oil is our calculation from block data reported by ADNSUR.

To Complete After the Trip

Our view. To be written once Dave is back.

Questions for the trip to answer:

- ☐ Is Argentina LNG financeable, and what follows the November 26 signing?
- ☐ When does Chevron's US$13.8bn RIGI application get approved, and what does it unlock?
- ☐ Is there any path to cutting the 8% export duty on shale crude?
- ☐ Does the VMOS first-cargo date of March or April 2027 hold?
- ☐ How much of the reform survives if Milei loses in 2027?

Lines to refresh before publishing:

- ☐ Argentina LNG: the outcome of the November 26 signing (Introduction, Key Points, LNG paragraph, ownership table)
- ☐ Macro: latest inflation and country risk
- ☐ Production: October data for Argentina and Neuquén (chart and Key Points)

Sources

Company third-quarter 2026 results, Argentine government data and trade press, as reported through November 11, 2026.

- September 2026 production record (Ámbito)
- RIGI approvals reach US$76bn (Infobae, November 9, 2026)
- RIGI extended to July 2027 (El Estratégico)
- Crude export duties (Infobae, January 29, 2026)
- Dividend rule of April 2025 (Beccar Varela)
- US$4bn of dividends remitted in 2026 (Diario Río Negro)
- Country risk above 900 (Infobae, November 7, 2026)
- Oldelval at its limit (Argenports)
- VMOS start-up schedule (Shale24)
- VMOS supertanker terminal (S&P Global)
- Basin capacity above 1.3 million barrels a day (Shale24)
- Infrastructure needed to 2030 (BBVA Research)
- Perito Moreno expansion (TGS)
- Southern Energy LNG (Offshore Technology)
- Argentina LNG joint development agreement (Eni)
- Argentina LNG RIGI filing (EconoJournal)
- Argentina LNG go-ahead date (Shale24)
- Shale resource rankings (EIA)
- Resource re-estimate (Infobae, August 13, 2026)
- Rig count (LM Neuquén)
- Chevron in Vaca Muerta (Reuters, October 23, 2026)
- Well costs (El Cronista)
- Productivity versus the Permian (Rystad)
- YPF Q3 2026 6-K
- Vista Q3 2026 call
- Pampa Q3 2026 call
- Harbour trading update, November 5, 2026
- Tenaris third frac fleet
- Halliburton and YPF completions contract
- H&P fiscal Q3 2026 results
- Neuquén 2027 budget (Diario Neuquino)
- TotalEnergies investment plans (Buenos Aires Times)
- Continental Resources and Phoenix (Mercuria)